

Birla Institute of Technology and Science-Pilani, Hyderabad Campus
Second Semester 2023-24
Midsemester Examination
Course No CS F364 Course Title: Design and Analysis of Algorithms
Type: Closed Time: 90 min Total Marks:90 (30%) Date: 14/3/24

Name: _____ ID: _____

Instruction: Attempt all the questions.

1. (5+2+3 points) 1. Prove $\sum_{i=1}^n i^k = \Theta(n^{k+1})$.
 2. Prove or disprove the following statement: $f(n) + g(n) = \Theta(\max(f(n), g(n)))$.
 3. Prove that if $f(n) = \Omega(g(n))$ then $f(n) \notin o(g(n))$.
2. (5+15 points) 1. Solve the following recurrence $T(n) = 2T(n/2) + n \log n$. [Hint: Master's Theorem does not work here].
 2. Solve the following recurrence using Generating Functions. Show all the steps.
 $a_n - 3a_{n-1} + 2a_{n-2} = 0$ where $a_0 = 1$ and $a_1 = 3$.
3. ((3+5+3+4) + (9+6) points) 1. (**MaxDiff**) Given an array of n elements, the problem is to find the maximum difference between any two elements.
 - A. A straightforward algorithm is to find the minimum and maximum value and compute the difference. Analyze the number of comparisons made in this algorithm.
 - B. Design a divide-and-conquer algorithm for solving the above problem with the number of comparisons approximately $\frac{3n}{2}$. Write the pseudocode of your algorithm and argue why your algorithm is correct in two to three sentences.
 - C. Write the recurrence relation of your algorithm.
 - D. Solve the recurrence to show that the number of comparisons made is approximately $\frac{3n}{2}$.
 2. Recall that a propositional formula in Conjunctive Normal Form (CNF) is one which is a conjunction (i.e. AND) of clauses, where each clause is a disjunction (i.e. OR) of literals, where each literal is either an atomic variable or its negation.
 - (a) Given a propositional formula in n variables in CNF, design a Divide-and-Conquer algorithm that outputs every satisfiable assignment of truth values to the atomic variables. Write clearly an English description of the Divide and Conquer phases of your algorithm. (Hint: Consider partitioning the variables).
 - (b) Write down recurrence relations for the worst-case and best-case running times of your algorithm and solve them.
4. (15+ (8+7) points) 1. Let's consider a long, quiet country road with houses scattered very sparsely along it. (We can picture the road as a long line segment, with an eastern endpoint and a western endpoint.) Further, let's suppose that despite the bucolic ¹ setting, the residents of all these houses are avid cell phone users. You want to place cell phone base stations at certain points along the road, so that every house is within four miles of one of the base stations. Give an efficient algorithm that achieves this goal, using as few base stations as possible and argue about its correctness. Note that a base station covers an interval of eight miles.
 2. (a) In the Traveling Salesperson Problem (TSP), a traveling salesperson wishes to travel to a set of n cities and return to the originating city, so as to minimize the total airfare. They are given the one-way airfare between every pair of cities. Briefly describe (in a few sentences) a greedy algorithm for TSP and prove that the algorithm is incorrect.
 - (b) In the knapsack problem, you are given n items, each of which has a value v_i and a weight w_i . You have a knapsack with capacity W . Briefly describe (in a few sentences) a greedy algorithm to fill the knapsack with items of maximum total value such that the total weight is at most W . Prove that the algorithm is incorrect.

¹relating to the pleasant aspects of the countryside and country life.